

Interactive K-Pop Re-Entry Analysis Dashboard for Music Chart Analytics Using Python, Pandas and Streamlit

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Abstract—The rapid growth of the K-Pop industry and digital music streaming platforms has resulted in the availability of large amounts of music chart data, creating opportunities for meaningful data analysis. This paper presents an Interactive K-Pop Re-Entry Analysis Dashboard that was developed to examine long-term chart performance and audience engagement using historical chart records. The application is implemented using Python, Pandas, Plotly, and Streamlit, providing an interactive environment for exploring various performance indicators such as Re-Entry Frequency, Comeback Momentum Score, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity Score. The collected data is processed and transformed into multiple analytical datasets, which are presented through interactive charts, summary tables, and visual performance metrics. The dashboard allows users to compare songs, identify chart trends, and gain insights into factors that influence long-term popularity within the K-Pop industry. To improve accessibility, the completed application is deployed on Streamlit Community Cloud, enabling users to access the dashboard through a web browser without additional software installation. The proposed system demonstrates how interactive visualization techniques can convert historical music chart data into valuable insights for researchers, music analysts, entertainment organizations, and K-Pop enthusiasts.

Keywords—K-Pop, Music Analytics, Data Visualization, Streamlit, Python, Pandas, Plotly, Dashboard, Interactive Analytics, Chart Re-Entry Analysis.

I. INTRODUCTION

The widespread adoption of digital music streaming services has changed the way people discover, access, and consume music across the world. Platforms such as **Spotify, YouTube Music, Apple Music, and Melon** continuously generate large volumes of music-related data, providing opportunities to analyze audience listening patterns and song performance. Among different music industries, **Korean Pop (K-Pop)** has experienced remarkable international growth over the last decade, making it an interesting domain for data analytics and visualization.

Music charts are commonly used to measure the popularity of songs, artists, and albums over time. Although many existing platforms provide information such as chart rankings and streaming statistics, they often do not capture important long-term performance indicators. Metrics such as **chart re-entry frequency, comeback momentum, retention duration, rank recovery, album comeback advantage, and fandom intensity** offer a deeper understanding of how songs

maintain or regain popularity after their initial release. Analyzing these factors can help reveal audience behavior, promotional effectiveness, and sustained fan engagement.

To explore these aspects, this work presents an **Interactive K-Pop Re-Entry Analysis Dashboard** developed using **Python, Pandas, Plotly, and Streamlit**. The dashboard processes historical chart data and converts it into interactive visualizations that enable users to examine different performance indicators through an intuitive web interface. Separate analytical modules have been designed to display **Re-Entry Frequency, Momentum Score, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity Score**, allowing users to compare songs and identify meaningful performance trends.

The application combines data preprocessing, statistical analysis, and visualization within a single platform, making complex chart information easier to understand. Interactive graphs, summary tables, and filtering options provide users with a practical way to explore historical chart performance without requiring advanced technical knowledge. The completed dashboard has been deployed using **Streamlit Community Cloud**, enabling online access through a standard web browser and improving the overall usability of the system.

The main objective of this project is to develop a user-friendly analytical dashboard that supports data-driven exploration of K-Pop music chart trends. By transforming raw chart data into meaningful visual insights, the proposed system can assist researchers, entertainment organizations, music analysts, and K-Pop enthusiasts in studying long-term song performance, audience engagement, and comeback patterns more effectively.

II. LITERATURE REVIEW

A. Existing Research on Music Analytics

The growing use of digital music platforms has encouraged researchers to explore different techniques for analyzing music-related data. Recent studies have applied data analytics, visualization methods, and machine learning algorithms to understand user listening behavior, streaming trends, and song popularity. These approaches have helped entertainment companies and artists gain valuable insights into audience preferences and overall music performance.

Several research works have focused on predicting the popularity of songs using features such as streaming statistics, listener interactions, and audio characteristics. Other studies have proposed interactive dashboards to simplify the visualization of large music datasets, allowing users to interpret analytical results more effectively. However, relatively few research efforts have specifically investigated how songs re-enter music charts after their initial release or how their popularity changes over an extended period.

B. Research Gap

Although considerable work has been carried out in music recommendation systems and popularity prediction, limited attention has been given to analyzing long-term chart performance. Important indicators such as **Re-Entry Frequency, Comeback Momentum, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity** are rarely studied together within a single analytical framework. In addition, many existing visualization systems provide only basic charts with limited interaction, making detailed exploration of historical chart trends difficult.

The proposed dashboard addresses these limitations by integrating multiple performance metrics into one interactive platform. Through dynamic visualizations and filtering options, users can examine different aspects of K-Pop chart performance and gain a more comprehensive understanding of long-term audience engagement.

C. Contribution of the Proposed Work

This research introduces an **Interactive K-Pop Re-Entry Analysis Dashboard** that combines multiple chart performance indicators within a unified analytical environment. Unlike conventional music dashboards that mainly present rankings or streaming statistics, the proposed system evaluates several long-term performance metrics, including Re-Entry Frequency, Momentum Score, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity Score.

The dashboard is implemented using **Python, Pandas, Plotly, and Streamlit**, providing an interactive web interface for exploring historical chart data. The application is further deployed using **Streamlit Community Cloud**, allowing users to access the dashboard through any standard web browser without requiring local software installation. This work demonstrates how interactive data visualization can improve the analysis of historical music chart behavior.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

The rapid expansion of digital music streaming services has led to the generation of extensive music chart datasets containing valuable information about song popularity and audience engagement. Despite the availability of this data, many existing analytical platforms mainly emphasize peak chart positions and streaming counts while providing limited support for studying long-term chart behavior. As a result, important performance indicators such as chart re-entry frequency, comeback momentum, retention duration, rank recovery speed, album comeback advantage, and fandom intensity are often overlooked.

This limitation makes it difficult for researchers, music analysts, and entertainment organizations to understand how songs maintain popularity over time or recover after leaving

the charts. Therefore, there is a need for an interactive analytical system capable of transforming historical music chart data into meaningful visual insights that support better understanding and informed decision-making.

B. Objectives

The main objectives of the proposed research are:

- To design and develop an interactive dashboard for analyzing historical K-Pop music chart performance.
- To preprocess historical chart datasets using Python and Pandas for reliable analysis.
- To compute key analytical metrics including Re-Entry Frequency, Momentum Score, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity Score.
- To visualize analytical results using interactive charts and tables developed with Plotly and Streamlit.
- To deploy the dashboard through Streamlit Community Cloud for convenient online access.
- To provide a simple and user-friendly interface that supports researchers, music analysts, and K-Pop enthusiasts.

C. Scope of the Study

This research focuses on the visualization and analysis of historical K-Pop music chart data through an interactive web dashboard. The proposed system is intended to examine song performance using multiple chart-based performance indicators rather than forecasting future rankings. Interactive charts and filtering features enable users to explore different dimensions of chart behavior and compare songs using several analytical metrics.

The scope of this study is limited to the historical datasets used during dashboard development. Real-time streaming information and live chart updates are beyond the scope of the current work. However, the framework can be extended in future by integrating live APIs, predictive machine learning models, and personalized recommendation features.

IV. PROPOSED METHODOLOGY

A. System Overview

The proposed **K-Pop Re-Entry Analysis Dashboard** is designed to provide an interactive platform for examining historical music chart performance. The system processes multiple analytical datasets representing different performance indicators such as Re-Entry Frequency, Momentum Score, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity. Before visualization, each dataset undergoes preprocessing and validation to ensure reliable analytical results.

The application is developed using **Python** for backend processing, **Pandas** for data manipulation, **Plotly Express** for interactive visualizations, and **Streamlit** for building the web interface. The completed dashboard is deployed using **Streamlit Community Cloud**, allowing users to access the application directly through a web browser.

B. Dataset Description

The dashboard utilizes multiple CSV files generated during the preprocessing stage. Each dataset corresponds to a specific analytical metric associated with K-Pop chart performance.

These datasets include:

- Re-Entry Frequency Dataset
- Momentum Score Dataset
- Retention Days Dataset
- Rank Recovery Speed Dataset
- Album Comeback Advantage Dataset
- Fandom Intensity Score Dataset

All processed CSV files are stored within the project directory and are loaded dynamically into the Streamlit application whenever users access the dashboard.

C. Data Preprocessing

Before generating visualizations, the collected datasets undergo several preprocessing operations to improve data quality and consistency. Missing values are addressed appropriately, duplicate entries are removed, irrelevant columns are eliminated, and data types are verified using Python and Pandas. After preprocessing, separate CSV files are created for each analytical metric to support efficient dashboard loading and visualization.

These preprocessing steps help ensure that the generated charts accurately represent the underlying music chart data and minimize inconsistencies during analysis.

D. Dashboard Development

The interactive dashboard is implemented using **Streamlit**, providing a responsive interface for exploring historical chart data. **Plotly Express** is used to generate interactive charts, while **Pandas** manages dataset loading and processing in the background.

The dashboard includes several analytical modules:

- Dashboard Overview
- Re-Entry Analysis
- Momentum Analysis
- Retention Analysis
- Rank Recovery Analysis
- Album Analysis
- Fandom Analysis

Users can also choose the number of records displayed through sidebar controls, enabling flexible exploration of different analytical results.

E. Deployment

After development and testing, the application is deployed using **Streamlit Community Cloud**, allowing users to access the dashboard through any modern web browser without installing additional software. The complete source code is maintained in a **GitHub repository**, providing version control and simplifying future updates and enhancements.

The deployment architecture consists of the GitHub repository, Streamlit Community Cloud, and the client-side web interface, ensuring reliable online accessibility across different devices. The overall workflow of the proposed system is illustrated in **Fig. 1**, which demonstrates the movement of data from preprocessing and KPI generation to interactive visualization and cloud deployment.

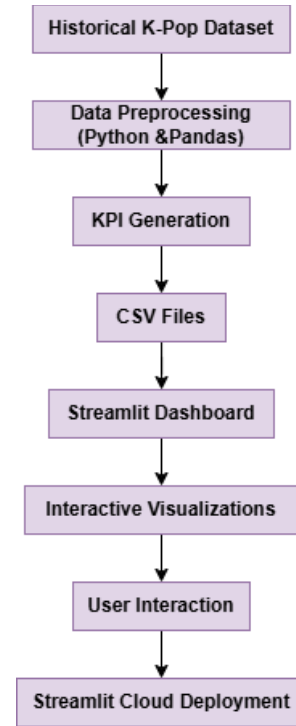


Fig. 1. Overall Architecture of the Proposed K-Pop Re-Entry Analysis Dashboard

F. System Workflow

The overall workflow of the proposed dashboard is illustrated in Fig. 2. The workflow begins with loading the processed CSV datasets, followed by data cleaning and feature engineering. Various Key Performance Indicators (KPIs) are then generated and visualized through interactive charts using Streamlit and Plotly. Finally, users can interact with the deployed dashboard to analyze K-Pop chart performance efficiently.

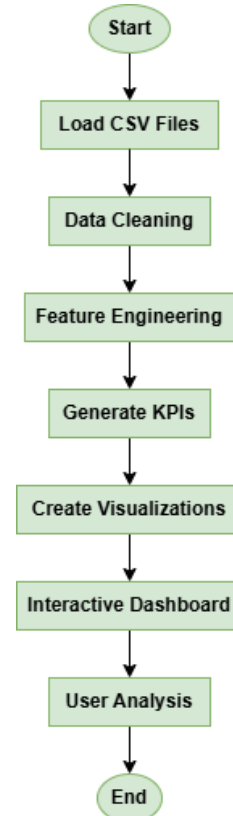


Fig. 2. Workflow of the Proposed K-Pop Re-Entry Analysis Dashboard.

G. Deployment Architecture

The deployment architecture of the proposed system is shown in Fig. 3. The project source code is managed using GitHub, while the dashboard is deployed through Streamlit Community Cloud. This setup enables users to access the application directly from any modern web browser, ensuring easy availability and convenient online access.

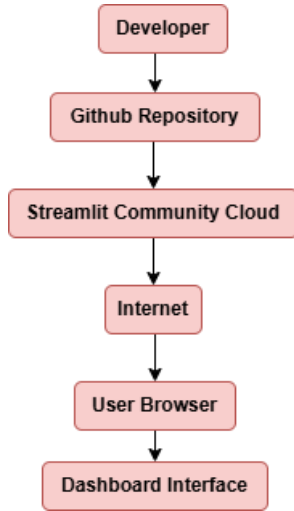


Fig. 3. Deployment Architecture of the Proposed Dashboard.

V. RESULTS AND DISCUSSION

A. Dashboard Home Page

The dashboard home page offers an overview of K-Pop chart performance through a simple and interactive interface. It presents key metrics such as the total number of songs analyzed, overall re-entry events, and the highest fandom score, along with highlights of the top-performing songs in each category. These summary insights provide users with a clear understanding of the dataset before they explore the detailed analytical modules available in the dashboard.

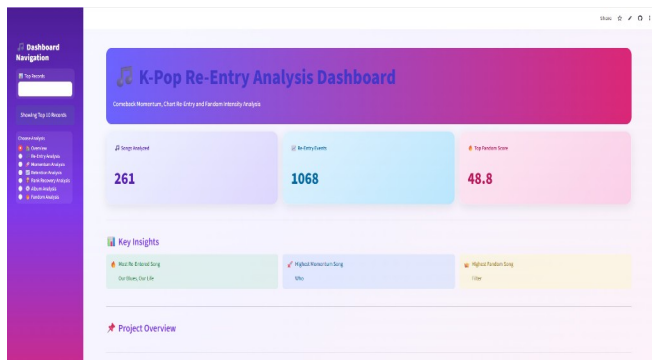


Fig. 4. Dashboard Home Page of the Proposed System

B. Re-Entry Analysis

The Re-Entry Analysis module displays how often songs return to the music charts after dropping from their previous positions. Using interactive bar charts, users can compare songs based on their re-entry frequency and identify tracks that maintain long-term popularity. This analysis provides useful insights into audience engagement and recurring chart performance over time.



Fig. 5. Re-Entry Frequency Analysis

C. Momentum Analysis

The Momentum Analysis module measures the comeback performance of songs based on their historical chart momentum scores. Interactive visualizations enable users to compare songs and identify releases that showed strong popularity growth after their debut. This analysis helps understand how songs regain attention and sustain audience interest over time.

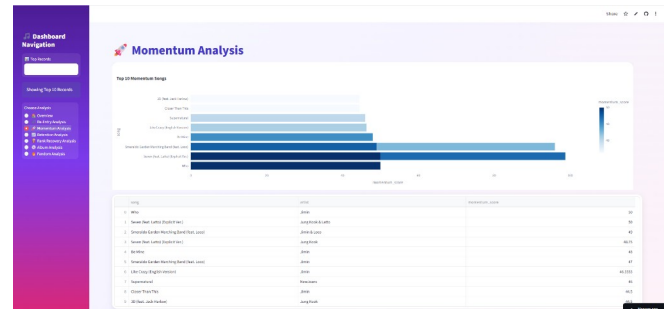


Fig. 6. Momentum Score Analysis

D. Retention Analysis

The Retention Analysis module examines how long songs remain on the music charts after their release. The interactive visualization ranks songs based on their retention period, helping users identify tracks that consistently maintain audience interest and long-term chart presence.

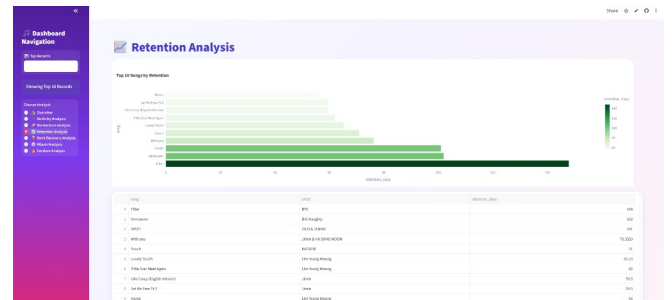


Fig. 7. Retention Days Analysis

E. Rank Recovery Analysis

The Rank Recovery Analysis module examines how effectively songs regain higher chart positions after a decline in rankings. The visualization helps users compare recovery patterns and understand the impact of audience engagement and promotional activities on a song's return to the charts.

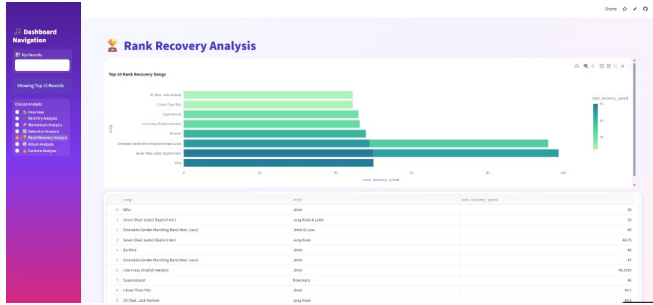


Fig. 8. Rank Recovery Speed Analysis

F. Album Analysis

The Album Analysis module compares the performance of different album release types using average popularity scores. This visualization helps users understand how various release formats influence chart performance and identify which type achieves stronger audience response over time.

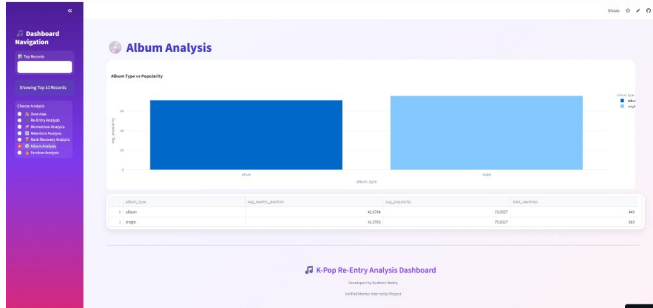


Fig. 9. Album Comeback Advantage Analysis

G. Fandom Analysis

The Fandom Analysis module evaluates audience loyalty using fandom intensity scores derived from historical chart data. Songs with higher scores generally demonstrate stronger fan support and better long-term chart performance. This visualization helps users understand the influence of dedicated fan communities on the sustained success of K-Pop songs.

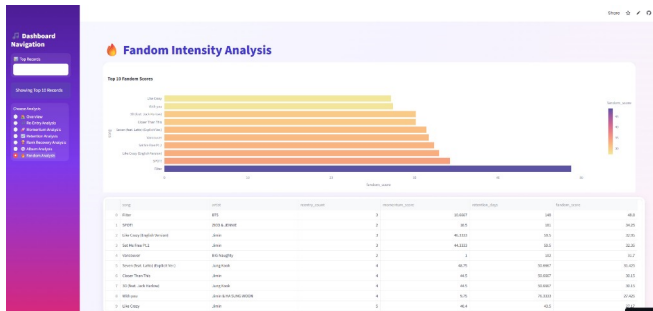


Fig. 10. Fandom Intensity Score Analysis

The proposed K-Pop Re-Entry Analysis Dashboard demonstrates an effective approach for exploring historical music chart performance through interactive data visualization. By combining multiple analytical metrics within a single platform, the dashboard enables users to better understand long-term chart trends, audience engagement, and song performance.

The dashboard analyzes multiple datasets to generate key performance indicators, including Re-Entry Frequency, Comeback Momentum, Retention Days, Rank Recovery Speed, Album Comeback Advantage, and Fandom Intensity. Built using Python, Pandas, Plotly, and Streamlit, the application provides an interactive interface that makes it easy to explore historical K-Pop chart trends and compare song performance.

Interactive charts and visual summaries help users interpret complex data more effectively. By deploying the application on Streamlit Community Cloud, the dashboard becomes easily accessible through any modern web browser without requiring additional software. Overall, the proposed system offers a practical platform for researchers, entertainment professionals, and K-Pop enthusiasts to study long-term chart performance and audience engagement.

VII. FUTURE SCOPE

The proposed system can be further enhanced through several improvements:

- Integration with real-time music streaming APIs for live chart updates.
- Implementation of machine learning models to predict future chart rankings.
- Inclusion of sentiment analysis using social media platforms such as Twitter and YouTube.
- Development of personalized music recommendation features based on user preferences.
- Expansion of the dashboard to support multiple international music charts.
- Incorporation of advanced analytics using deep learning techniques.
- Addition of mobile-responsive features for improved accessibility across devices.
- Integration of artist performance comparison and genre-based analytics.

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